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Eco-46

Dummy Variable.

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Dummy Vari. Answer

Dummy Variable Trap :-

Let, consider a model

$$Y_i = \beta_1 + \beta_2 D_{2i} + \beta_3 D_{3i} + u_i$$

Here, public school teachers salaries are classified by three regions. - 1) Northeast and North central

2) South

3) West, ^{intercept}

Y_i = average salary of public school teachers in district i

$D_{2i} = 1$ if the state is in North west/north central

$D_{2i} = 0$ otherwise

$D_{3i} = 1$ if the state is in south
 $= 0$ otherwise.

If a qualitative variable has m categories, introduce only $(m-1)$ dummy variables. In our example since the qualitative variable region has three categories we introduce only two dummies. If we do not follow the rule we fall into what is called the dummy variable trap that is the situation of perfect collinearity or perfect multicollinearity.

Overcome Dummy variable trap:

By introducing as many dummy variables as the number of categories of that variable provided we don't introduce the intercept term in such a model. That is

$$Y_i = \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + u_i \quad \text{--- (2)}$$

We do not fall into dummy variable trap, as there is no longer perfect collinearity. Here,

β_1 = Mean salary in the west

β_2 = Mean salary in the Northeast and North central.

β_3 = mean salary in the south.

In other words with the intercept term suppressed and allowing a dummy variable for each category we obtain directly the mean values of the various categories.

ANCOVA MODELS:

Regression models containing an admixture of quantitative and qualitative variables are called analysis of covariance (ANCOVA) models. ANCOVA models are an extension of the (ANOVA) models in that they provide a method of statistically controlling the effects of quantitative regressors, called co-variables or control variables, in a model that includes both quantitative and qualitative or dummy, regressors.

Reconsider the previous example, maintaining that the average salary of public school teachers may not be different in the three regions if we take into account any variables that can't be standardized across the regions. Consider the variable expenditure by public on public schools by local authorities. To see if this is the case, we develop the following model :-

$$Y_i = \beta_1 + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 D_{2i} X_i + U_i$$

Y_i = average annual salary of public school.

X_i = spending on public school per pupil.

$D_{2i} = 1$, if the state is in the Northeast / N.E.
 $= 0$ otherwise

$D_{3i} = 1$ if state is in the south,
 $= 0$ otherwise.

Here, we treat the west as the benchmark category.
 and the quantitative variable X which in the context of the ANCOVA models is known as a covariate.

The Dummy Variable Alternative to the Chow Test :

We could not tell whether the difference in the two regressions was because of differences in the intercept terms or the slope co-efficients or both. Very often this knowledge itself is very useful.

For time period 1: $Y_t = \lambda_1 + \lambda_2 X_t + U_{1t}$ — (I)

For time period 2: $Y_t = \gamma_1 + \gamma_2 X_t + U_{2t}$ — (II)

time period in total: $Y_t = \alpha_1 + \alpha_2 X_t + U_t$ — (III)

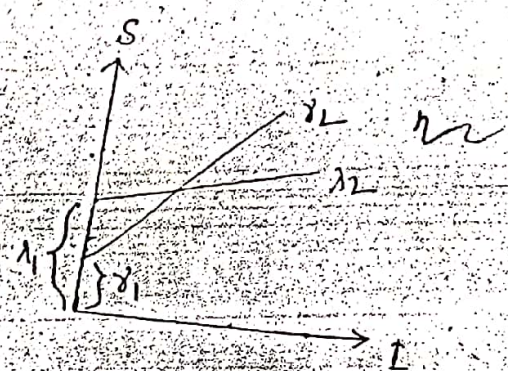
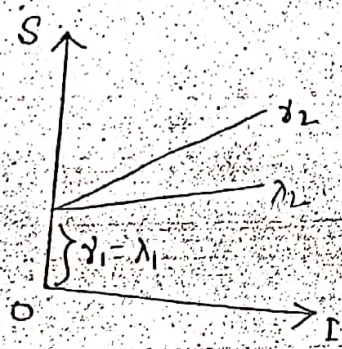
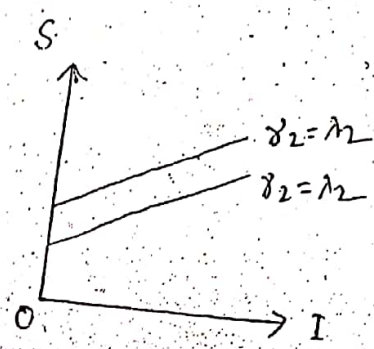
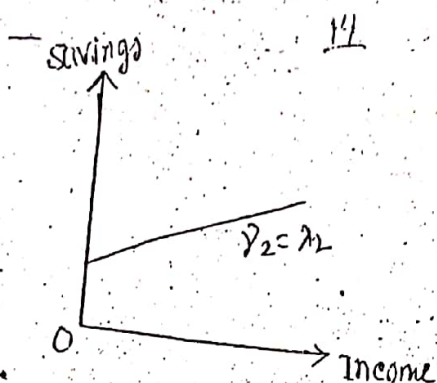
Referring to equation (I) and (II) we see that there are four possibilities which we illustrate in the following figure

1. Both the intercept and the slope co-efficients are the same in the two regressions. This, the case of coincident regression. (Fig - A)

2. Only the intercepts in the two regressions are different but the slopes are the same. This is the case of parallel regressions which is shown in (Fig - B)

3. The intercepts in the two regressions are the same but the slopes are different. This is the situation of concurrent regression.

4. Both the intercepts and slopes in the two regressions are different. This is the case of dissimilar regressions.



The multistep chow test procedure tells us only if two regressions are different without telling us what the source of the difference is. The source of difference if any can be pinned down by pooling all the observations and running just one multiple regression as shown below:

$$y_t = \alpha_1 + \alpha_2 D_t + \beta_1 X_t + \beta_2 (D_t X_t) + u_t \quad \text{--- (1)}$$

y = savings X = income t = time
 $D = 1$ for period 2
 $= 0$ otherwise

To see the implications of Eq (4) and assuming $E(u_i) = 0$

Mean saving function for period 1

$$E(Y_t | D_t = 0, X_t) = \alpha_1 + \beta_1 X_t \quad - (5)$$

Mean saving function for period 2

$$E(Y_t | D_t = 1, X_t) = (\alpha_1 + \alpha_2) + (\beta_1 + \beta_2) X_t \quad - (6)$$

These are the same functions as equation (1) and equation (2) with $\lambda_1 = \alpha_1$ and $\lambda_2 = \alpha_2$; $\gamma_1 = (\alpha_1 + \alpha_2)$ and $\gamma_2 = (\beta_1 + \beta_2)$. Therefore, estimating Equation (5) is equivalent to estimating the two individual savings functions in equation (1) and (2).

α_2 = differential slope intercept

β_2 = differential slope coefficient / slope drifter.

indicating by how much the slope coefficient of the second period's savings function differs from that of first period.

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Multicollinearity is essentially a sample (regression) phenomenon:

Ans: Mc is essentially a sample phenomenon in the sense that even if the X variables are not linearly related in the population they may be so related in the particular sample at hand.

When we postulated the theoretical or population regression function (PRF) we believe that all the X variables included in the model have a separate or independent influence on the dependent Y. But it may happen that in any given sample that is used to test the PRF some or all of the X variables are so highly collinear that we can't isolate their individual influence on Y.

So, our sample lets us down, although the theory says that all the X's are important. In short our sample may not be rich enough to accommodate all X variables in the analysis.

$$\text{Consumption} = \beta_1 + \beta_2 \text{Income}_i + \beta_3 \text{Wealth}_i + u_i$$

Here, besides income wealth of the consumer is also an important determinant of consumption.

Now it may happen that when we obtain data on income & wealth the two variables may be highly if not perfectly correlated. Wealthier people generally tend to have higher incomes. Thus, although in theory income and wealth are logical candidates to explain the behavior of consumption expenditure in practice it may be difficult to disentangle the separate influence of income and wealth on consumption expenditure.

Underlying Assumption¹⁷ of the "Durbin-Watson" test :

1. The explanatory variables the x 's are nonstochastic or fixed in repeated sampling.
2. The regression model includes the intercept term. If it is not present, as in the case of the regression through the origin it is essentially to return the regression including the intercept term to obtain the RSS.
3. The disturbances u_t are generated by the first order autoregressive scheme $u_t = \rho u_{t-1} + e_t$. Therefore it can't be used to detect higher order autoregressive schemes.
4. The error term is used to be normally distributed.
5. The regression model does not include the lagged values of the dependent variable as one of the explanatory variables. Thus the test is inapplicable in models of the following type:
$$Y_t = \beta_1 + \beta_2 x_{2t} + \beta_3 x_{3t} + \dots + \beta_K x_{Kt} + \gamma Y_{t-1} + u_t$$

where, Y_{t-1} = one period lagged value of Y . Such models are known as autoregressive models.
6. There are no missing observations in the data.

OLS Estimation Disregarding Heteroscedasticity :-

Let, $Y_i = \beta_1 + \beta_2 X_i + u_i$

The OLS estimator of $\beta_2 \Rightarrow \hat{\beta}_2 = \frac{\sum x_i y_i}{\sum x_i^2}$

$$= \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2} \quad \text{--- (1)}$$

now, variance is now given by the following expression.

$$\text{Var}(\hat{\beta}_2) = \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2} \quad \text{--- (2)}$$

which is obviously different from the usual variance formula obtained under the assumption of homoscedasticity, namely

$$\text{Var}(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_i^2} \quad \text{--- (3)}$$

Here, $\sigma_i^2 = \sigma^2$

If we estimate disregarding Heteroscedasticity, then the situation become serious. Again if we not only use $\hat{\beta}_2$ but also continue to use the usual variance formula given in (3) even if heteroscedasticity is present or suspected. This is the more likely case of too we discuss how because running a standard OLS and ignoring heteroscedasticity will yield variance of $\hat{\beta}_2$ as given in Eq (3). First of all, $\text{var}(\hat{\beta}_2)$ given in Eq (3) is a biased estimation of $\text{var}(\hat{\beta}_2)$ given in (2). That is on the average it overestimate or underestimate the latter and in general we can't tell whether the bias is positive/negative because it depends on the nature of the relationship between σ_i^2 and the values taken by explanatory variable X .

The bias arises from the fact that $\hat{\sigma}^2$, the conventional estimator of σ^2 namely $\sum \hat{u}_i^2 / (n-2)$ is no longer an unbiased estimator of the latter when heteroscedasticity is present. As a result we can no longer rely on the conventionally computed confidence intervals and the conventionally employed t and F tests.

In short, if we persist in using testing procedures despite heteroscedasticity whatever conclusions we draw or inferences we make may be very misleading.

The most striking feature of these results is that OLS with or without correction for heteroscedasticity, consistently overestimates the true standard error obtained by the (correct) GLS procedure especially for large values of α , thus establishing the superiority of GLS. These results also show that if we do not use GLS and rely on OLS allowing for or not allowing for heteroscedasticity

In the presence of heteroscedasticity we have to use GLS. or can take corrective action such as weighted least squares regression or some other technique.

Remedial Measure :-

The remedy depends on the knowledge one has about the nature of interdependence among the disturbances that knowledge about the structure of autocorrelation :-

Let, two period variable regression model

$$Y_t = \beta_1 + \beta_2 X_t + u_t \quad \text{--- (1)}$$

and assume that the error term follows the AR(1) scheme, namely

$$u_t = \rho u_{t-1} + \epsilon_t \quad -1 < \rho < 1 \quad \text{--- (2)}$$

When ρ is known :-

If the co-efficient of first order autocorrelation is known, the problem of autocorrelation can be easily solved. If eq (2) holds true at time t , it also holds true at time $(t-1)$. Hence

$$Y_{t-1} = \beta_1 + \beta_2 X_{t-1} + u_{t-1} \quad \text{--- (3)}$$

Multiplying Eq (3) by ρ on both sides, we obtain

$$\rho Y_{t-1} = \rho \beta_1 + \rho \beta_2 X_{t-1} + \rho u_{t-1} \quad \text{--- (4)}$$

subtracting (4) from (3) gives

$$(Y_t - \rho Y_{t-1}) = \beta_1 (1 - \rho) + \beta_2 (X_t - \rho X_{t-1}) + \epsilon_t \quad \text{--- (5)}$$

Here, $\alpha \cdot \epsilon_t = \rho u_{t-1} - u_{t-1}$

We can express Equation (5) as

$$Y_t^* = \beta_1^* + \beta_2^* X_t^* + \epsilon_t \quad \text{--- (6)}$$

Here, $\beta_1^* = \beta_1 (1 - \rho)$

$Y_t^* = (Y_t - \rho Y_{t-1})$

$X_t^* = (X_t - \rho X_{t-1})$

$\beta_2^* = \beta_2$

since the error term in equation (6) satisfies the usual OLS assumptions, we can apply OLS to the transformed variables y^* and x^* and obtain estimators with all the optimum properties namely BLUE. Equation 6 is tantamount to using generalized least squares (GLS).

Equation 5 is known as the generalized or quasi difference equation. It involves regressing y on x not in the original form but in the difference form, which is obtained by subtracting in the previous period to current time period. If this case we lose the one observation. To avoid this loss of one observation the first observation on y and x is transformed as follows: $y_1\sqrt{1-\rho^2}$ and $x_1\sqrt{1-\rho^2}$. This transformation is known as Prais-Winsten transformation.

Multicollinearity

$$\hat{\beta}_2 = \frac{(\sum y_i x_{2i})(\sum x_{3i}^2) - (\sum y_i x_{3i})(\sum x_{2i} x_{3i})}{(\sum x_{2i}^2)(\sum x_{3i}^2) - (\sum x_{2i} x_{3i})^2} \quad x_{3i} = \lambda x_{2i}$$

$$\hat{\alpha} = (\hat{\beta}_2 + \lambda \hat{\beta}_3) = \sum x_{2i} y_i / \sum x_{2i}^2$$

$$\text{var}(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_{2i}^2 (1 - r_{23}^2)}$$

$$\text{cov}(\hat{\beta}_2, \hat{\beta}_3) = \frac{-r_{23} \sigma^2}{(1 - r_{23}^2) \sqrt{\sum x_{2i}^2 \sum x_{3i}^2}}$$

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